Fiscal Year:	FY 2013	**Task Last Updated:**	FY 02/12/2013
PI Name:	Trivedi, Rohit Ph.D.		
Project Title:	Dynamical Selection of Three-Dimensional Interfacial Patterns in Directional Solidification		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rohit.k.trivedi@gmail.com	**Fax:**	FY 515-294-4291
PI Organization Type:	UNIVERSITY	**Phone:**	515-450-1948
Organization Name:	Iowa State University		
PI Address 1:	Dept. of Materials Science and Engineering		
PI Address 2:	100 Wilhelm Hall		
PI Web Page:			
City:	Ames	**State:**	IA
Zip Code:	50011	**Congressional District:**	4
Comments:	Other email: trivedi@ameslab.gov		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	06/01/2007	**End Date:**	11/30/2012
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Strutzenberg Littles, Louise	**Contact Phone:**	(256) 544-0946
Contact Email:	louise.s@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date is now 11/30/2012 (Ed., 6/22/2012) NOTE: End date changed to 5/31/2012 (from 5/31/2010) per NSSC info dtd 9/2009 (10/2009) NOTE: Received NCE to 5/31/2010 (from 5/31/2009) per NSSC info dtd 6/2009 (10/2009)		
Key Personnel Changes/Previous PI:	Dr. Jiho Gu left the research group at the end of 2011.		
COI Name (Institution):	Karma, Alain (Northeastern University)		
Grant/Contract No.:	NNX07AK69G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed research consists of experiments on the International Space Station on the dynamics of three-dimensional interface pattern evolution in directional solidification, and the supporting critical ground-based research and the rigorous modeling to analyze observations from the low gravity experiments. Low gravity experiments were successful in that no effect of convection was observed. Initial results from the space station were analyzed to characterize the dynamical evolution of spatially extended patterns. Several new results were observed that include the continuous variation in primary spacing with velocity over the cell-dendrite growth regime, and first observations of the formation of multiplets in three-dimensions and the presence of novel oscillating modes during the growth of cellular array at low velocities. A rigorous theoretical model, using the phase-field approach was developed that gave excellent agreement with the experimental observation on the complex oscillating modes observed experimentally. Detailed experimental studies were also carried out terrestrially in a thin sample in which diffusive growth conditions are present to examine the variation in primary spacing with velocity and to establish some critical characteristics of multiplets observed in low gravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of fundamental quantitative knowledge of microstructure evolution under dynamic growth conditions is crucially important for improving large scale industrial processes from casting to welding. The results will provide basis to predict and design reliable processing conditions that optimize microstructures and hence material properties and performance. These results are crucial in the design of high precision castings for aerospace and defense industries. This research provides for the first time the benchmark results on three-dimensional spatial arrangement of cells and dendrites since convection effects strongly influence the patterns and prevent the fundamental physics that govern these patterns under diffusive growth conditions. The benchmark data obtained under low gravity conditions will be a cornerstone for the validation of phase-field approach through unique quantitative comparison of model predictions and benchmark experiments in three dimensions. These models can then be used successfully in the design of casting parameters that are crucial for the production of high quality precision castings by U.S. industries. The knowledge gained from this study will also be applied to other pattern forming phenomena that occur in different fields of science such as combustion, fluid physics, cryo-biology, biology, and food science.
Task Progress:	Results from experiments on International Space Station. Detailed experimental studies in a microgravity environment where (i) convection is reduced sufficiently to obtain growth under diffusive conditions and (ii) the interface evolution can be directly visualized in three dimensions provided an unprecedented opportunity to obtain quantitative benchmark data that can be compared to predictions of computational models. Directional solidification experiments were thus carried out on the International Space Station by using the DECLIC-DSI facility built by the French Space Agency CNES. A total of 39 experiments were carried out over a period of 100 days, and they were carried out for two different thermal gradient values and several growth rates in an alloy of succinonitrile-0.24 wt% camphor. Since the patterns evolve under dynamical growth conditions, the evolution process depends on the detailed sequence through which the final velocity is reached. These experiments were carried out at a fixed velocity and with a change in velocity so that the different dynamics can be compared. All experiments were highly successful in that no convection effects were observed and the three-dimensional pattern was uniform across the entire cross-section. A large amount of data still needs to be analyzed since we received the images from the International Space Station only recently. These results will be analyzed to reconstruct the three-dimensional evolution of cellular and dendritic arrays and to characterize the effect of the change in velocity on the spatial arrangement of the arrays. Ground-based experiments in thin samples. Several experiments in thin samples were carried out to obtain an insight into the growth pattern that evolves under diffusive conditions. Two sets of experiments were carried out: (1) to compare the results in three dimensions obtained from ISS with the results in a thin sample and, (2) to examine the characteristics of multiplets in thin samples.
Bibliography Type:	Description: (Last Updated: 04/03/2024)
Articles in Peer-reviewed Journals	Mota FL, Bergeon N, Tourret D, Karma A, Trivedi R, Billia B. "Initial transient behavior in directional solidification of a bulk transparent model alloy in a cylinder." Acta Materialia. 2015 Feb 15;85:362-77. https://doi.org/10.1016/j.actamat.2014.11.024 , Feb-2015